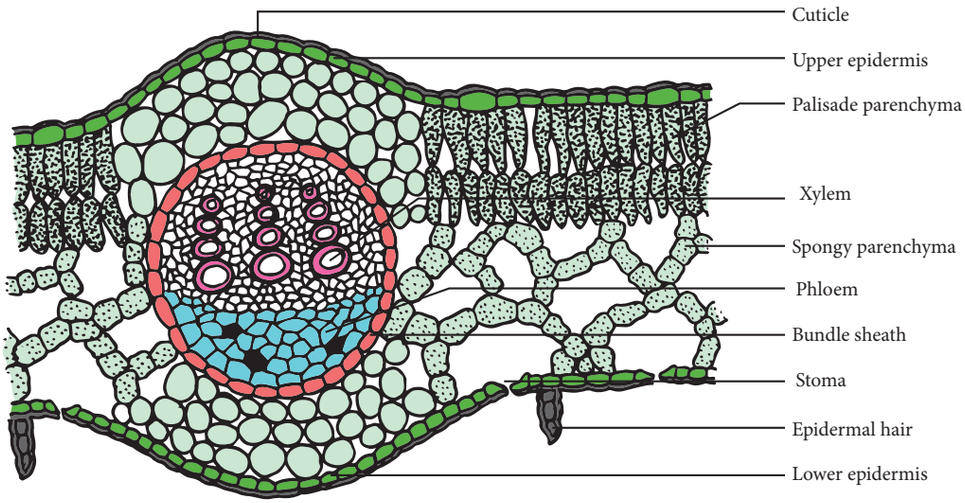


Figure 12.6 Transverse section of Dicot leaf

(ii) **Lower epidermis:** It is a single layer of parenchymatous cells with a thin cuticle. It contains numerous stomata. Chloroplasts are present only in guard cells. The lower epidermis helps in the exchange of gases. The loss of water vapour is facilitated through this chamber.

(iii) **Mesophyll:** The tissue present between the upper and lower epidermis is called mesophyll. It is differentiated into Palisade parenchyma and Spongy parenchyma.

a) **Palisade parenchyma:** It is found just below the upper epidermis. The cells are elongated. These cells have more number of chloroplasts. The cells do not have intercellular spaces and they take part in photosynthesis.

b) **Spongy parenchyma:** It is found below the palisade parenchyma tissue. Cells are almost spherical or oval and are irregularly arranged. Cells have intercellular spaces. It helps in gaseous exchange.

(iv) **Vascular bundles:** Vascular bundle are present in mid-rib and lateral veins. Vascular bundles are conjoint, collateral and closed. Each vascular bundle is surrounded by a sheath of parenchymatous cells called **bundle sheath**. Each vascular bundle consists of xylem lying towards the upper epidermis and phloem towards the lower epidermis.

12.8 Internal Structure of Monocot Leaf (Isobilateral Leaf) Ex.Grass Leaf

The transverse section of a monocot leaf reveals the following structures.

(i) **Epidermis:** Monocot leaf has upper and lower epidermis. Epidermis is made up of parenchyma cells. Cuticle is present on the outer wall stomata are present on both upper and lower epidermis. Some cells of upper epidermis are large and thin walled they are known as **bulliform cells**.

(ii) **Mesophyll:** It is the ground tissue that is present between both epidermal layers. Mesophyll is not differentiated into palisade and spongy parenchyma. The cells are irregularly arranged with inter-cellular spaces. These cells contain chloroplasts.

(iii) **Vascular bundles:** Large number of vascular bundles are present, some of which are small and some are large.

10th_Science_Unit-12.indd 179 11-01-2020 15:36:12